

RAY: Resilience Assistant for You

An Accountable GeoAI System for Place-Based Disaster Intelligence

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ABSTRACT

RAY (Resilience Assistant for You) is a disaster-resilience GeoAI system designed around a simple requirement: every public claim must stay within the evidence and leave a record that can be checked. Hourly federal feeds and event builders turn hazard, damage, vulnerability, population, and imagery data into H3 products for three deep cases. A *Steward Harness* checks spatial outputs, records SHA-256 lineage, enforces separate claim and distribution policies, and tracks whether evidence is retained, re-derivable, or cited-only. Residents receive context and guidance rather than parcel damage assessments; planners choose the trade-off between damage and social vulnerability. Six open vision-language models were evaluated on 1,225 labelled samples per model, but their outputs remain evaluation products and are excluded from operational claims. Three recorded failures led to new publication, citation, and evidence-scope controls. The result is an executable accountability design for autonomous geospatial workflows, with explicit limits on coverage, disclosure, and evidence use.

CCS CONCEPTS

- Information systems → Geographic information systems;
- Computing methodologies → Artificial intelligence.

KEYWORDS

GeoAI agents, accountability, disaster resilience, social vulnerability, provenance, policy enforcement, vision-language models, H3

1 INTRODUCTION

Large-language-model agents can now operate geographic information systems with little human intervention [4, 7]. In disaster response that capability meets an unforgiving audience: an emergency manager acting on a tile-level damage estimate, or a resident

reading a sentence about their own street, cannot audit a transformer. The prevailing answer is a disclaimer, which locates accountability in the reader. RAY locates it in the system. Following Yang and Zou [6], we treat an agent’s claim as valid only when the *outcome* passes executable spatial checks, the *process* is preserved and inspectable, and the claim is *institutionally* authorized — the right actor asserting the right thing at the right resolution — and we add a fourth question raised by third-party data: what can a *reader* do to check this, and may the evidence be kept at all?

This paper describes the deployed system¹ along the four axes of the Track A rubric. *Autonomy* (§2): the pipelines write, run and check spatial analysis on their own, and the places where a human sets a parameter are deliberate. *Robustness and generalizability* (§3): missing data, failing endpoints and mismatched geographies are declared rather than smoothed, by one harness that produced three cases in two states and two hazard types. *Social good alignment* (§4): vulnerability, exposure and priority are defined operationally, the trade-off is the user’s, and equity is enforced as rules about who may be told what. *Innovation and reflection* (§5–6): model output treated as governed evidence, three self-caught violations, and the limits of the whole approach.

2 AUTONOMY UNDER CONSTRAINT

What runs without a human. The data plane is a set of GitHub Actions pipelines. Four keyless federal connectors (USGS earthquakes, NWS alerts, NHC tropical cyclones, and NIFC/WFIGS wildfires) refresh hourly and write append-only snapshots plus a national watch product. Deep-case builders ingest registered files, execute the spatial logic, validate the outputs, and append audit rows without manual low-level cleaning. Table 1 summarizes the committed products and the limits that travel with them. A hashed dataset registry (134,272 files, ~33 GB) carries a label crosswalk onto one severity scale. CDC’s –999 sentinel becomes a counted missing value, never a number. When the Hugging Face Milton pairs disagreed with the Figshare table on 182 of 2,556 pairs, the builder used the table as truth and recorded the discrepancy.

Debugging as a first-class record. Autonomous code fails. The harness’s answer is not a retry loop but fail-closed stages whose failures are written down: an over-strict tract-join assertion

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¹Live site, app, and repository: <https://rayford295.github.io/ray-resilience/>

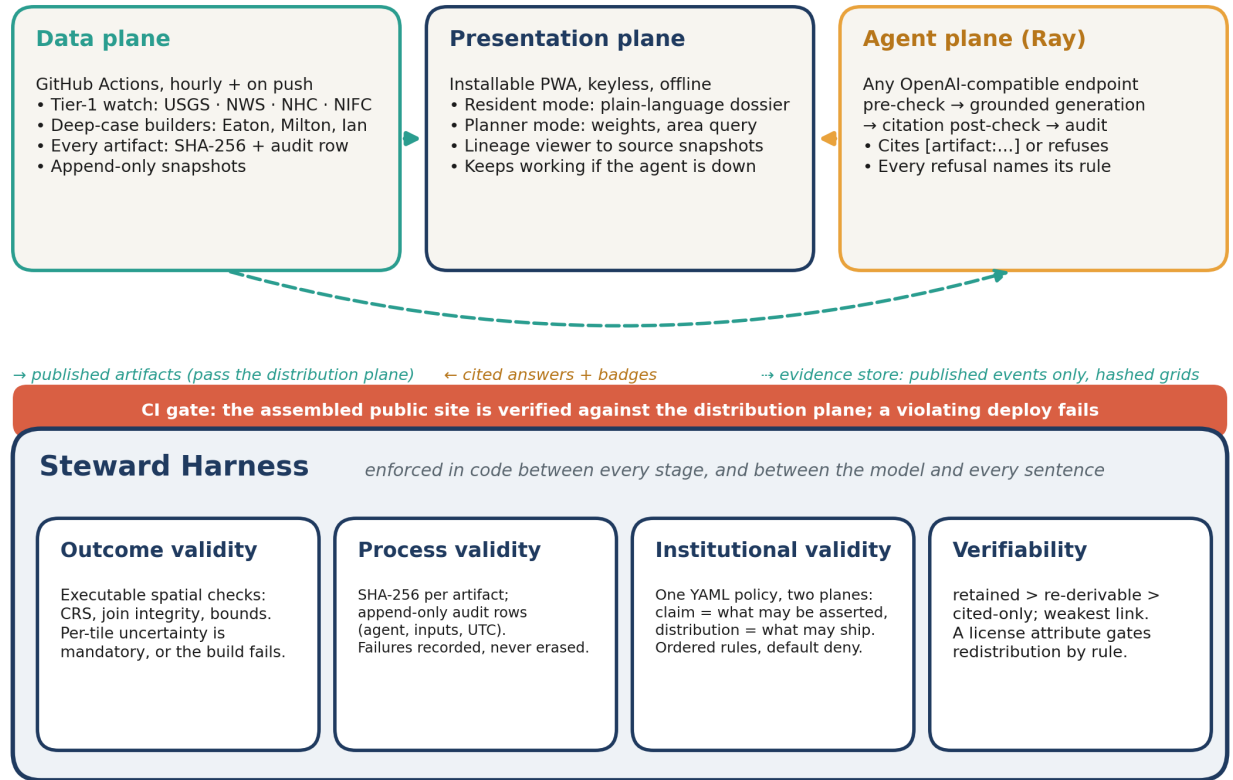


Figure 1: RAY system architecture. The claim plane governs what the agent may say; the distribution plane governs what the public build may ship. Both use the same policy source and default to denial when no rule applies.

Table 1: Deep-case products and declared limits. Population is 2020 Census exposure; facilities are OpenStreetMap locations, not operational-status claims.

case	committed result	limit carried forward
Eaton 2025	18,428 DINS points → 265 cells; 2,244 cross-view samples → 109; 46,341 residents; 27 facilities	40 inaccessible points; repairable $n=30$; tract-to-cell SVI is approximate
Milton 2024	2,556 image pairs → 15 cells; 5,618 debris cells; 772,293 residents; 200 facilities	season-cumulative imagery cannot isolate Milton damage
Ian 2022	886 matched samples → 190 cells; 4,121 positions → 413 density cells; 5,428 residents; 79 facilities	density cells support coverage, not point-level severity

during the Eaton SVI build was rejected, logged, and corrected in a later run, and the log keeps both rows. Every artifact carries a SHA-256 digest and an audit row (agent, inputs, UTC time); every grid cell must carry an uncertainty block or the build fails.

Where the human is, on purpose. The language-model agent, Ray, answers questions; it does not choose which pipelines run. We declined an LLM task planner deliberately: a model deciding which analyses execute cannot be reconciled with a policy that decides what may be asserted. The human-in-the-loop points are explicit.

A user selects a role (resident or planner), which changes what the agent may say. A planner sets the one value-laden parameter the system computes with — an inspection priority $t \cdot \text{damage} + (1 - t) \cdot \text{SVI}$ on a live slider, tiles scored with partial data flagged, every adjustment audit-logged — and can draw an area and ask about it, receiving an answer that covers only the evaluated tiles inside it and highlights the tiles it cited. A resident enters an address and receives one of three answers: covered, outside the evaluated areas, or *not determined* when a layer could not be read. The system will not claim an address is outside areas it failed to look at.

3 ROBUSTNESS AND GENERALIZABILITY

The Steward Harness sits between every pipeline stage, and between the model and every sentence it emits. *Outcome validity*: CRS assertions, join integrity, sanity bounds, mandatory per-tile uncertainty. *Process validity*: SHA-256 lineage in an append-only audit the app renders from any layer back to source snapshots; the gateway’s audit stores a point as its H3 cell and a question as a digest, so a caller can prove their question produced a row without the log retaining what a resident typed about their own home. *Institutional validity*: one YAML policy with two planes of ordered, first-match rules, default deny, validated at load. The *claim plane* scopes

Table 2: Operational definitions used by RAY.

concept	definition and guardrail
Vulnerability	CDC/ATSDR SVI 2022 tract percentiles; cell display is marked as downscaled
Exposure	2020 Census block population assigned by centroid; pre-event population only
Damage	official DINS classes, a declared repairability mapping, or human perception labels; never a model guess
Priority	$t \cdot \text{damage} + (1 - t) \cdot \text{SVI}$; the planner chooses t and the system logs it
Coverage	covered, outside evaluated areas, or not determined; a failed layer is not "outside"

what may be asserted by role, evidence tier, resolution and geography; the *distribution plane* scopes what a build may *publish*, and CI verifies the assembled site against it and fails the deploy on violation. One `published_events` list feeds both planes. *Citation by default*: every factual sentence cites an artifact identifier or is refused; the exemption set is closed, so an unanticipated sentence form costs a refusal, not an uncited claim. *Verifiability*: each artifact carries retained > re-derivable > cited-only with weakest-link aggregation and a license attribute; where map-platform terms forbid retention, the harness attests to the request plus a response digest, a record publishable because it holds no content, and a property test proves no payload reaches the audit.

Real-world complications, declared. *Missing data*: 42 of 295 Palisades images lack EXIF GPS, so the grid covers photographed structures and says so; a source that fails leaves the watch badge reporting *mapped-of-total* and naming the source; a class with $n=30$ is declared as lacking power. *Endpoints*: an ArcGIS error envelope fails the wildfire connector closed instead of reporting zero fires; live adapters time out at 30–60 s and record the timeout; a model call that times out is a recorded `LLMUnavailable`, and a model the serving stack cannot load is a declared failure, not a zero. *Spatial misalignment*: tract-level SVI on an H3 cell is a downscaling approximation stated per feature; a census block straddling tiles is wholly assigned to its centroid’s tile, and no areal interpolation is invented; population outside the evaluated tiles is reported as a total, never mapped into cells the event has no evidence for; post-event Milton imagery is season-cumulative (Debby, Helene, Milton) and every feature carries that attribution caveat.

Generalizability. The same harness, policy and app produced the Eaton Fire 2025 (CA, wildfire), Hurricane Milton 2024 (FL) and Hurricane Ian 2022 (FL) cases; adding an event is a dossier directory and one entry in `published_events`, and the policy contains no geography. The Tier-1 watch is nationwide by construction. Where evidence does not generalize the system says so; Table 1 separates verified severity from density-only coverage. The test suite is the robustness evidence: 363 Python and 67 app tests in CI, including a cell-by-cell policy-matrix sweep (every role $\times$ tier $\times$ resolution), adversarial claim checks, connector tests against real error envelopes, and an allowlist drift check that fails on any hand edit. Judges can run it in two commands from the README.

4 SOCIAL GOOD ALIGNMENT

Operational definitions. RAY uses established measures and keeps the remaining choices visible (Table 2). Each definition is a committed artifact with a hashed source snapshot, so a reader can inspect both the measure and its limits.

Equity as enforced rules. Fairness here is not a metric but a set of institutional constraints about who may be told what, each

a named rule. *No parcel for anyone*: no role receives a parcel-level claim; tile evidence never becomes a statement about a house. *No damage assessment for residents*: they receive exposure context and guidance, not raw damage assessments of their own street, because a false “not significantly affected” costs the reader more than a refusal. *Planner damage at Tier 3 only*: planners see tile-level damage where Tier-3 cross-view evidence exists. Declared unknowns render as prominently as findings; *not determined* is a distinct answer from *outside*; and no rule authorizes facility claims, so answers about hospitals and shelters default-deny rather than guess.

Community perspectives, honestly. The Milton truth labels are human damage perception, not an instrument reading, and the resident mode speaks in plain language with every qualifier attached. Beyond that we claim little: no community co-designed these definitions, and no local knowledge enters the pipeline yet. What the architecture offers is the *place* where negotiated definitions would go — a declarative, human-readable policy file in which a community’s answer to “who may be told what” is a rule, not a prompt — and the record that would let a community check whether the system kept its word.

5 MODEL OUTPUT AS GOVERNED EVIDENCE

The obvious next step for an imagery-rich system is to let a vision-language model (VLM) grade damage. The harness turns that step into a question: under what conditions may a model’s guess become evidence? Model output is a distinct class, `model_derived`, that carries its own measured accuracy or carries nothing; no claim-plane rule admits it; the gateway’s evidence store refuses to load it. We ported the Damage Recognition prompts of RAPID [5] verbatim (Prompt C single image, five DINS classes; Prompt B pre/post pair, three human-perception classes), recorded a SHA-256 prompt digest in every prediction, and served open weights with Ollama on a 24 GB RTX 3090, so the run is keyless and repeatable. Three cases: Palisades 2025 (RAPID Dataset C2, 295 images, all graded); Milton 2024 (100 pairs per class, seed 2026); Eaton 2025 (300 per class, the $n=30$ minority class fully included; the post-event image only, declared; five classes collapsed onto the set’s three repairability classes). Every prediction is a record with image sha256, truth, prediction, response digest, latency and model digest; an unparseable answer is recorded as such, never coerced. A stage aborts if more than 20 % of answers are off-schema, a metric leaves $[0, 1]$, fewer than 95 % of graded samples fall inside the event’s committed grid, or any product lacks an uncertainty block. We report accuracy and RAPID’s NCSE = $|y - \hat{y}| / (K - 1)$.

What the runs say. Table 3: (i) on a single post-event image every open model beats the 7B reference and the best is within two points of the paper’s closed models, with no monotone size effect — gemma3:12b (0.546) is within a point of qwen3-vl:32b (0.554); (ii) on pre/post pairs no open model beats the 7B reference, and each fails with a different fixed bias — qwen3-vl:32b calls 258 of 300 pairs Severe (Mild recall 0.01), gemma3:12b calls 262 Moderate, mistral 213 — so a vote across models would not be a fix; (iii) Eaton’s $n=30$ class is weak for every model (recall 0.03–0.43) while the large classes

Table 3: Six open VLMs and a 7B reference, one pass at temperature 0, quantised weights, same seeded samples and prompt digests. Accuracy / NCSE; columns are not comparable with each other. The closed-model row is RAPID’s published number (same Palisades images; a 150-pair Milton subset).

model	Palisades (5)	Milton pairs (3)	Eaton (3)
qwen2.5vl:7b (ref.)	0.475 / 0.213	0.597 / 0.238	0.910 / 0.049
qwen3-vl:32b [†]	0.554 / 0.176	0.381 / 0.431	0.901 / 0.053
qwen2.5vl:32b ^{†‡}	0.507 / 0.206	0.560 / 0.240	0.931 / 0.039
gemma3:27b	0.529 / 0.187	0.500 / 0.257	0.914 / 0.049
mistral-small3.2:24b	0.505 / 0.203	0.487 / 0.258	0.935 / 0.037
gemma3:12b	0.546 / 0.178	0.427 / 0.287	0.906 / 0.051
qwen3-vl:8b	0.514 / 0.195	0.462 / 0.324	0.924 / 0.041
closed (RAPID)	0.573 GPT-5-mini	0.591 GPT-5.1	—

[†]8k context so the weights fit the card; [‡]95% in VRAM, recorded, because a spilled model is a different run. A seventh model (llama3.2-vision:11b) could not be loaded by the serving stack and is declared a serving failure, not scored.

Table 4: Recorded failures and resulting controls.

failure	change and verification
Parcel-level DINS file reached the public site	added a distribution plane, generated allowlist, and CI gate; public files fell from 30 to 16 and the restricted URL returned 404
Uncited reassurance passed the post-check	replaced the numeric heuristic with citation-by-default; acceptance changed from 9/13 to 7/13 because the two uncited reassurances were refused
Gateway loaded every event on disk	load only policy-listed events, exclude model-derived grids, record every exclusion, and refuse unrestricted scans

sit above 0.9; (iv) the qwen3 family spends its time in hidden reasoning before a 60-character answer, and on Milton 24 of qwen3-vl:8b’s pairs hit the client timeout, recorded as unanswered (8.3%, under the bound).

What the harness did with it. the agreement grids sit beside the human-labelled grids they are evaluated against, flagged model_derived; the publication planner ships them as an evaluation; the claim plane has no rule that admits them; the gateway skips them and records the exclusion. A reader can verify every number from the records without a GPU. What no one can do — and what the system therefore does not do — is turn a 0.93 into a sentence about a street.

6 REFLECTION

Three incidents changed the architecture. Table 4 records the gap, the fix, and the check. The repeated lesson is practical: a constraint written in one place and enforced in another is enforced nowhere.

Limitations. The builders read a ~33 GB corpus on one workstation, so third parties can verify the committed artifacts and audit logs but not regenerate them. The VLM numbers are one pass on 4-bit weights on a 24 GB card; Milton and Eaton are seeded samples, and the closed-model Milton reference used a different subset. The gateway runs locally without a hosted endpoint, auth or rate limiting, so the public demo ships without chat; the re-derivable regime has run only against a stub. Three deep cases are not national analysis capability, and the app says so.

Ethics and the societal role of the agent. The disclosure risk in this domain is concrete: per-structure damage is a statement about a family’s home, and our first incident shows how easily it leaks without a rule that names it. We therefore treat the resolution cap, the role denials and the content-free audit not as features but as the conditions under which an agent may speak to the public at all. The value judgements — which index counts as vulnerability, how damage trades against it — are handed to people, in a file they can read and a slider they can move, and the system records that they did. Autonomous-GIS agents [4, 7] and operational platforms such as TDIS [3] show what such systems can compute; registries (EM-DAT [1], GDACS [2]) show what they can record. What none carries, per row, is what resolution of claim the evidence authorizes and whether a reader may keep it. Our position is that an agent in public decision-making should be a steward of evidence rather than an oracle: it should compute more than it is allowed to say, say only what it can cite, and leave a record — including of its own failures — that the public it serves can check.

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